
Concept Questions: Chapter 3

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ME-3640 – Heat Transfer

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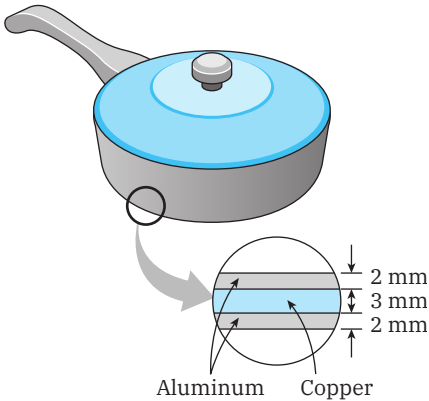


Abstract

Fundamental principals and concepts are far more important to engineering expertise than understanding of example systems. These concept questions (shamelessly lifted from the textbook) will help you test your knowledge after reading the chapter summary and building a concept map. Use the relative ease and difficulty you experience while answering these questions to prioritize your efforts for the remainder of this chapter.

1 Steady Heat Conduction in Plane Walls

- 3-1C** Consider heat conduction through a wall of thickness L and area A . Under what conditions will the temperature distributions in the wall be a straight line?
- 3-2C** Consider heat conduction through a plane wall. Does the energy content of the wall change during steady heat conduction? How about during transient conduction? Explain.
- 3-3C** What does the thermal resistance of a medium represent?
- 3-4C** Can we define the convection resistance for a unit surface area as the inverse of the convection heat transfer coefficient?
- 3-5C** Consider steady heat transfer through the wall of a room in winter. The convection heat transfer coefficient at the outer surface of the wall is three times that of the inner surface as a result of the winds. On which surface of the wall do you think the temperature will be closer to the surrounding air temperature? Explain.
- 3-6C** How is the combined heat transfer coefficient defined? What convenience does it offer in heat transfer calculations?
- 3-7C** Why are the convection and the radiation resistances at a surface in parallel instead of being in series?
- 3-8C** Consider steady one-dimensional heat transfer through a plane wall exposed to convection from both sides to environments at known temperatures T_1 and T_2 with known heat transfer coefficients h_1 and h_2 . Once the rate of heat transfer Q has been evaluated, explain how you would determine the temperature of each surface.
- 3-9C** Someone comments that a microwave oven can be viewed as a conventional oven with zero convection resistance at the surface of the food. Is this an accurate statement?
- 3-10C** Consider two cold canned drinks, one wrapped in a blanket and the other placed on a table in the same room. Which drink will warm up faster?
- 3-11C** The bottom of a pan is made of a 4mm-thick aluminum layer. In order to increase the rate of heat transfer through the bottom of the pan, someone proposes a design for the bottom that consists of a 3mm-thick copper layer sandwiched between two 2mm-thick aluminum layers. Will the new design conduct heat better? Explain. Assume perfect contact between the layers.



8mm-thick glass sheet under identical conditions.

3-12C Consider a surface of area A at which the convection and radiation heat transfer coefficients are h_{conv} and h_{rad} , respectively. Explain how you would determine (a) the single equivalent heat transfer coefficient, and (b) the equivalent thermal resistance. Assume the medium and the surrounding surfaces are at the same temperature.

3-13C How does the thermal resistance network associated with a single-layer plane wall differ from the one associated with a five-layer composite wall?

3-14C Consider steady one-dimensional heat transfer through a multilayer medium. If the rate of heat transfer Q is known, explain how you would determine the temperature drop across each layer.

3-15C Consider a window glass consisting of two 4mm-thick glass sheets pressed tightly against each other. Compare the heat transfer rate through this window with that of one consisting of a single

2 Thermal Contact Resistance

- 3-43C** What is thermal contact resistance? How is it related to thermal contact conductance?
- 3-44C** Will the thermal contact resistance be greater for smooth or rough plain surfaces?
- 3-45C** Explain how the thermal contact resistance can be minimized.
- 3-46C** A wall consists of two layers of insulation pressed against each other. Do we need to be concerned about the thermal contact resistance at the interface in a heat transfer analysis or can we just ignore it?
- 3-47C** A plate consists of two thin metal layers pressed against each other. Do we need to be concerned about the thermal contact resistance at the interface in a heat transfer analysis or can we just ignore it?
- 3-48C** Consider two surfaces pressed against each other. Now the air at the interface is evacuated. Will the thermal contact resistance at the interface increase or decrease as a result?

3 Generalized Thermal Resistance Networks

- 3-58C** What are the two approaches used in the development of the thermal resistance network for two-dimensional problems?
- 3-59C** The thermal resistance networks can also be used approximately for multidimensional problems. For what kind of multidimensional problems will the thermal resistance approach give adequate results?
- 3-60C** When plotting the thermal resistance network associated with a heat transfer problem, explain when two resistances are in series and when they are in parallel.

4 Heat Conduction in Cylinders and Spheres

3-73C What is an infinitely long cylinder? When is it proper to treat an actual cylinder as being infinitely long, and when is it not?

3-74C Can the thermal resistance concept be used for a solid cylinder or sphere in steady operation? Explain.

3-75C Consider a short cylinder whose top and bottom surfaces are insulated. The cylinder is initially at a uniform temperature T_i and is subjected to convection from its side surface to a medium at temperature T , with a heat transfer coefficient of h . Is the heat transfer in this short cylinder one- or two-dimensional? Explain.

5 Critical Radius of Insulation

- 3-96C** What is the critical radius of insulation? How is it defined for a cylindrical layer?
- 3-97C** Consider an insulated pipe exposed to the atmosphere. Will the critical radius of insulation be greater on calm days or on windy days? Why?
- 3-98C** A pipe is insulated to reduce the heat loss from it. However, measurements indicate that the rate of heat loss has increased instead of decreasing. Can the measurements be right?
- 3-99C** Consider a pipe at a constant temperature whose radius is greater than the critical radius of insulation. Someone claims that the rate of heat loss from the pipe has increased when some insulation is added to the pipe. Is this claim valid?
- 3-100C** A pipe is insulated such that the outer radius of the insulation is less than the critical radius. Now the insulation is taken off. Will the rate of heat transfer from the pipe increase or decrease for the same pipe surface temperature?

6 Heat Transfer from Finned Surfaces

3-105C Hot air is to be cooled as it is forced to flow through the tubes exposed to atmospheric air. Fins are to be added in order to enhance heat transfer. Would you recommend attaching the fins inside or outside the tubes? Why? When would you recommend attaching fins both inside and outside the tubes?

3-106C What is the reason for the widespread use of fins on surfaces?

3-107C What is the difference between the fin effectiveness and the fin efficiency?

3-108C The fins attached to a surface are determined to have an effectiveness of 0.9. Do you think the rate of heat transfer from the surface has increased or decreased as a result of the addition of these fins?

3-109C Explain how the fins enhance heat transfer from a surface. Also, explain how the addition of fins may actually decrease heat transfer from a surface.

3-110C How does the overall effectiveness of a finned surface differ from the effectiveness of a single fin?

3-111C Hot water is to be cooled as it flows through the tubes exposed to atmospheric air. Fins are to be attached in order to enhance heat transfer. Would you recommend attaching the fins inside or outside the tubes? Why?

3-112C Consider two finned surfaces that are identical except that the fins on the first surface are formed by casting or extrusion, whereas they are attached to the second surface afterwards by welding or tight fitting. For which case do you think the fins will provide greater enhancement in heat transfer? Explain.

3-113C The heat transfer surface area of a fin is equal to the sum of all surfaces of the fin exposed to the surrounding medium, including the surface area of the fin tip. Under what conditions can we neglect heat transfer from the fin tip?

3-114C Does the (a) efficiency and (b) effectiveness of a fin increase or decrease as the fin length is increased?

3-115C Two pin fins are identical, except that the diameter of one of them is twice the diameter of the other. For which fin is the (a) fin effectiveness and (b) fin efficiency higher? Explain.

3-116C Two plate fins of constant rectangular cross section are identical, except that the thickness of one of them is twice the thickness of the other. For which fin is the (a) fin effectiveness and (b) fin efficiency higher? Explain.

3-117C Two finned surfaces are identical, except that the convection heat transfer coefficient of one of them is twice that of the other. For which finned surface is the (a) fin effectiveness and (b) fin efficiency higher? Explain.

A Rubrics

Submissions of this assignment will be graded upon the criteria found in this section. Review these carefully, and complete your work so that application of this rubric produces your desired score.

 Criterion 1 Problems	
<i>Homework is completed with appropriate effort, neatness, etc.</i>	
Points	Description
20	Honest effort to participate and learn: every problem shows an answer or questions which need resolution.
10	Rushed or haphazard work: some problems show no effort at all.
0	Work is absent, wholly incorrect, or plagiarized.